

CSI247 Data Structures

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Semester I 2022/23

Course Outline

Objectives: learn and implement common data structures.

Reinforce the concept that there are costs and benefits for every data structure.

Topics will include but not limited to the following:

- Revise Object oriented programming
- Search and sorting algorithms using arrays
- Recursion
- Data Structures
 - Stacks, queues, trees, graphs, hash tables (using arrays and linked lists)

Schedule

- **3 Lectures per week**
 - MWF, all at 07:00 – 08:00
 - @ 252/001
- **1 Lab session per week**
 - R, 7 – 9, 9-11 or 11-1pm
 - Pick1, and stick to it throughout the semester
 - @ 247-293/294/295/296/297
 - @232-119/120
- **Class Material**
 - Text book: Data Structures & Algorithms in Java by Goodrich, Tamassia and Goldwasser.
 - Lecture notes and videos: see Moodle

Assessment

- CA: 50%
 - Labs (20%)
 - 2 Tests (15%):
 - T1: 11th September 2022
 - T2: 30th October 2022
 - 2 Lab tests (15%)
 - LT1: 1st September 2022
 - LT2: 20th October 2022
- Final Exam: (50%)